ETHICAL AI · CONSOLIDATED SUMMARY NOTES

Unit 1 — Introduction to AI & Ethics, and Core Ethical Principles

Strong & weak AI • Descriptive / Normative / Meta ethics • Ethics & law • Machine ethics • Moral diversity & testing • Bias & fairness, transparency & explainability, privacy & security, robustness & reliability

Contents

1. What is AI? (definitions, ML, robots)
2. Strong vs weak AI & types of AI systems
3. Types of ethics
4. Relationship between ethics and law
5. Machine ethics & examples
6. Moral diversity and testing
7. Why responsible AI & the core principles
8. Bias and fairness
9. Transparency and explainability
10. Privacy and security
11. Robustness and reliability
12. Quick revision & likely questions

Sources: **Ethical AI** (Bartneck, Lütge, Wagner, Welsh, Springer 2021) — Ch. 2 *What Is AI?* and Ch. 3 *What Is Ethics?*; **Introduction to Responsible AI** (Manure, Bengani, Saravanan, Apress 2023) — Ch. 1 *Introduction* (core ethical principles).

1. What is AI?

Who	Definition
Kaplan & Haenlein	A system's ability to **correctly interpret external data, learn from it, and use that learning to achieve goals** through flexible adaptation.
Poole & Mackworth	Study of computational **agents that act intelligently**: actions fit circumstances & goals, flexible, learn from experience, choose well under limits.
Russell & Norvig	Study of agents that receive **percepts** and take **actions** — each agent is a function mapping percepts → actions.

- **Four schools** (Russell & Norvig): machines that **think like humans**, **act like humans**, **think rationally**, **act rationally**.
- Common core: designing **intelligent agents that achieve goals**. AI is a *moving target* — "AI is whatever computers can't do yet".
- **Autonomy** is context-dependent (Greek *auto-nomos*, "self-rule"): engineering → works without an operator; bioethics → patient's right to decide; Kantian ethics → choosing one's own moral rules.

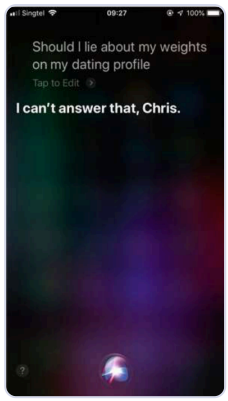


Figure 2.1 — Siri: "I can't answer that, Chris." (Should I lie about my weight on my dating profile?)

Why this "simple" question is hard

Needs speech recognition → language understanding (homophone *lie*) → world knowledge (dating, honesty) → a **moral stance**: **consequentialist** (maximise everyone's happiness), **deontological** (lying violates duty, Kant), **virtue** (be honest, Aristotle) — plus business risk for Apple.

1.1 Machine learning

Type	Idea	Example
Supervised	Learn from **labelled** data (classification/regression) → produces a *classifier*. Evaluated by **true-positive** and **false-positive** rate on a separate **test set**.	Spam filter
Unsupervised	Find structure without labels (clustering, PCA); exploratory.	Customer segments
Reinforcement	Learn a **policy** (action → expected reward) from a reward signal.	Robot navigating to a goal

Ethics link

Supervised labels are assigned by humans, so **societal bias can be baked into the classifier** — the thread that leads into Unit 2.

1.2 Robots & why AI is hard

- **Robot** = AI that is **situated** (real world) and **embodied** (physical body). Works by **Sense** → **Plan** → **Act**; an error at one step propagates to the next.
- **System integration** (sensors/actuators with different failure modes) is one of the hardest parts.
- **Hard for AI:** limited sensing; **poor generalisation** (face recogniser fails on a profile view); **no common sense / common ground**; no innate knowledge (a robot is a "blank slate"); repeated **hype cycles**.
- **Science vs fiction:** HAL, Terminator, Westworld inflate expectations (the "Frankenstein complex"). **Sophia** (Saudi "citizenship", 2017) is scripted PR — no learning, no adaptation.

2. Strong vs Weak AI & Types of AI Systems

Turing Test (1950)

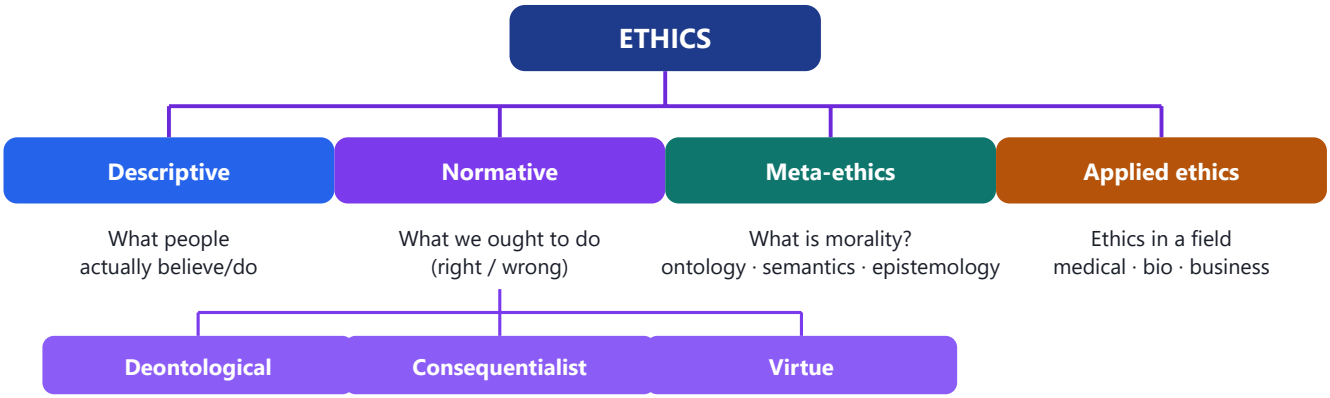
If a human interviewer cannot tell a machine from a person in conversation, the machine is judged intelligent. Replaces the vague word "thinking" with a testable task. *Criticism:* no time limit specified; passing is not clearly necessary/sufficient for intelligence.

	Weak (narrow) AI	Strong AI (Searle, 1980)
Scope	One narrowly defined task	General intelligence across many problems
Claim	Simulates intelligent behaviour	A suitably programmed computer *has a mind* like humans
Status	**All current AI** (chess/Go engines, loan scoring, Siri)	**Not yet achieved**

Types of AI systems: Expert systems (rule-based; e.g. mortgage advisors) • **Planning systems** (e.g. SPIKE for the Hubble telescope) • **Computer vision** (object recognition) • **Machine learning**.

3. Types of Ethics

Ethics = theory of morality (principles, general norms); **morality** = the actual rules, values and norms guiding action. Kant: ethics asks "*What should I do?*" (Greek *ethos* = custom; Cicero's *mores* → "moral").



Taxonomy of ethics (Ch. 3)

Branch	Question	Key points
Descriptive	What do people actually believe and do?	Empirical — moral psychology, experimental economics. E.g. the **ultimatum game**: people give up profit for fairness. Gives input to normative ethics ("experimental ethics").
Normative	What *ought* to be done?	Prescriptive; claims general validity ("stealing is wrong for everybody"). Three theories below.
Meta-ethics	What is morality itself?	**Ontology** (what has moral worth), **semantics** (meaning of right/wrong/ought), **epistemology** (how moral truths are known). "Theory of ethics".
Applied	Ethics in concrete fields	Medicine, biotech, business. Influence runs **both ways**: practice shapes theory (solidarity means sharing in a family, fair competition in society).

3.1 The three normative theories

Theory	Judges an action by	Detail	CSR example
Deontological (duty ethics)	The action itself — intention, duty, rule	**Kant's categorical imperative**: act only on a maxim you can will as a **universal law** without contradiction.	Critics ask about the company's **motive** (PR vs genuine)

Consequentialist	Foreseeable **consequences**	Bentham & Mill's **utilitarianism** — maximise happiness.	Motive irrelevant; only the **impact** on society counts
Virtue ethics	The agent's **character**	Plato's cardinal virtues (**wisdom, justice, fortitude, temperance**); Aristotle extends to 11 moral + intellectual virtues.	Is the firm acting honestly and justly?

4. Relationship Between Ethics and Law

Popular view: "*ethics starts where the law ends*" — the book challenges it:

1. **Laws have an ethical side** — anti-pollution and anti-trust laws are also ethical norms.
2. **Ethics works as "soft law"** — firms follow ethical standards the law doesn't demand, to protect reputation and share price (e.g. *fair-trade* coffee as a selling point); the effect is almost like **hard law**.

> **Remember**
> Law = enforceable minimum; ethics = broader, often precedes and shapes law (e.g. hate-speech laws on social media). They overlap but are not opposites.

5. Machine Ethics

> **Machine ethics**
> Asks: **what would it take to build an ethical AI that makes moral decisions?** Machines lack **phenomenology** (feelings, consciousness) and moral intuition — they only process *data about* feelings. Critics (van Wynsberghe & Robbins) say roboticists haven't given strong reasons to build moral robots.

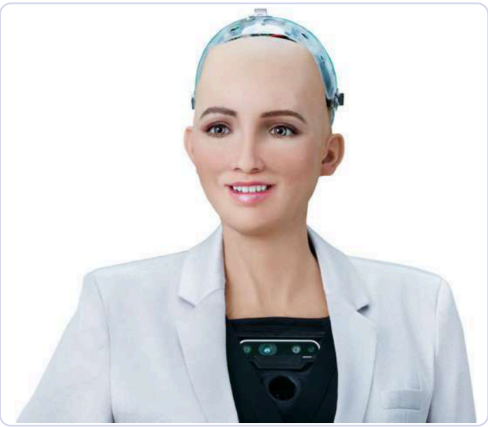


Figure 3.3 — Sophia: lifelike, but no feelings or consciousness

Sensors → Symbol grounding → Moral cognition (logic) → Action

- **Symbol grounding** = converting raw sensor data into symbols (objects, events, properties, relations).
- **Logic** = truth-preserving inference (Socrates is a man; all men are mortal ⇒ Socrates is mortal).

5.1 Machine ethics examples

> **Easy case — speeding-ticket robot**
> Rule: *if driver X is speeding then the robot is obligated to issue a ticket to X.* Fact: X is speeding ⇒ issue ticket. Clear rule + clear symbols = easy.

Hard case — toddler vs letter

A robot on its way to post a letter sees a toddler fall into a 1 m stream. Two duties clash: **rescue** vs **post**. The robot has no empathy or sense of urgency, so it needs **causal understanding** and a **utility scale**: toddler = +1,000,000, on-time letter = +1 ⇒ rescue.

When naive utilitarian arithmetic fails

Now the robot drives a truck with **1,000,001 letters** (+1 each) ⇒ 1,000,001 > 1,000,000 ⇒ it lets the toddler drown. Simple utility sums give **counter-intuitive** results; implementing deontology needs a principled way to **resolve clashes of duties**.

6. Moral Diversity and Testing

- **Core problem: no agreement on the correct moral theory** — how do we implement moral competence in AIs and robots if we have no moral theory to inform the design?
- **Approach — test cases:** design many ethical test cases; moral competence = ability to pass them. Iterating on new cases may expand competence and give insight into theory.
- **Testing & certifying fairness (active research):** IEEE standards project on **algorithmic bias** (2017); **AI Fairness 360**, **audit-AI**; services like O'Neil Risk Consulting & Algorithmic Auditing; Facebook's **Fairness Flow**.
- Also: codes of ethics for robotics engineers and HRI professionals. Some researchers remain **pessimistic** about machine morality.

7. Why Responsible AI & the Core Ethical Principles

(Manure et al., Ch. 1) AI promises large gains (autonomous vehicles, precision agriculture, personalised learning, NLP, diagnostics) but its power to **make decisions, learn and act autonomously** creates an ethical dilemma. **Responsible AI** is the guiding framework for AI's **development, deployment and governance**: aligning innovation with societal values through ethical principles, accountability and transparency across the AI lifecycle.

Why it matters	Explanation
Mitigating bias & discrimination	Models trained on biased data reproduce prejudice; responsible AI demands rigorous data preprocessing, algorithmic transparency and a pursuit of fairness.
Privacy vs innovation	AI's appetite for data threatens individual privacy; needs strong encryption, secure storage and access control so people keep agency over their data.
Human-centric design	AI should **augment** humans, not replace them; keep **human oversight** and develop **explainable AI** so decisions are comprehensible and traceable.
Governance	Robust frameworks, codes of conduct and regulation across legal, social and cultural boundaries; averts harm and ensures AI serves the collective good.

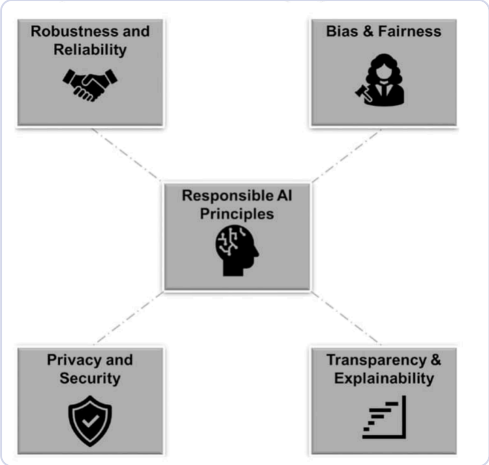


Figure 1-1 — The four core principles of Responsible AI (Manure et al.)

Bias & Fairness • Transparency & Explainability • Privacy & Security • Robustness & Reliability

8. Bias and Fairness

Bias & fairness

Bias seeps into AI through the data used to train it — algorithms learn patterns from vast datasets that already contain biases of human decisions and social structures → discriminatory outcomes, stereotypes, widened disparities. **Fairness** = treating all individuals equitably regardless of background, demographics or characteristics.

- Eliminating *all* bias may be infeasible — the focus is on **mitigating** glaring biases that lead to unjust or harmful outcomes.
- **Fairness is an ethical imperative** (a "north star"): goes beyond statistical definitions to moral and societal obligations, redressing historical and systemic inequalities.
- **Algorithmic fairness:** decisions must not depend on **sensitive attributes** (race, gender, socioeconomic status); fairness **metrics, algorithms and techniques** exist to assess and rectify bias.
- **Challenge — quantifying fairness:** no universally accepted definition; competing notions (equal treatment vs redressing historical disparities); trade-offs between fairness types and with accuracy.
- **Mitigation path:** awareness & education; data preprocessing (**re-sampling, re-weighting**); **adversarial debiasing**; diversity in data collection, model development and evaluation.
- **Societal impact:** biased AI perpetuates inequality *and* erodes trust in technology — organisations should adopt AI-ethics frameworks across the development lifecycle.

Continued in Unit 2

Types of bias, real cases, detection & mitigation techniques and a Python implementation.

9. Transparency and Explainability

	Meaning
Transparency	Openness and comprehensibility of an AI system's decision-making — illuminates the **"black box"** (how inputs become outputs). Fosters **accountability**, lets people identify and challenge errors or unintended consequences, and builds **trust**.

Explainability	Bridges the **cognitive gap** between complex non-linear AI processing and human cognition — answers "*Why did it make this decision?*" and "*How did it arrive at this conclusion?*" by turning outputs into explanations people can follow.

- **Applications: finance** (loan decisions), **health care** (diagnoses, treatment recommendations), **criminal justice** (stops predictive models perpetuating racial/socioeconomic bias). Also supports **auditing** and **regulatory compliance**.
- **Challenges:** (1) trade-off between **model complexity** and interpretability (deep nets are hard to explain); (2) dynamic models that update/adapt need explanations that stay meaningful through the lifecycle; (3) **proprietary** concerns — balancing transparency with intellectual property and the public's right to transparency.

10. Privacy and Security

- **Privacy** is a cornerstone of personal freedom; AI accumulates and analyses huge amounts of data, so personal information must be handled with the highest level of consideration. **Informed consent:** people have the right to know how their data is used and shared.
- **Data minimisation:** collect, process and retain only the data essential for a specific AI task — lowers the risk of unintended exposure and breaches.
- **Data security:** protect data from unauthorised access, manipulation or theft — **encryption, access controls, secure storage**, with protection built into every stage from collection to disposal.
- **Challenges:** (a) AI needs vast data, so sophisticated **anonymisation** must keep individuals' privacy intact in aggregated data; (b) global **dataflows** need harmonised privacy/security regulation and international standards.
- **Solutions: privacy-preserving AI** — **federated learning** (learn from decentralised data) and **homomorphic encryption** (compute on encrypted data).
- Privacy and security are "inseparable twins": strong protection builds trust and prevents misuse of AI for surveillance.

11. Robustness and Reliability

	Meaning
Robustness	Capacity of algorithms to stay **effective and accurate amid complexity, uncertainty and adversarial scenarios** — generalise to novel situations, data variations or deliberate attempts to deceive. E.g. a self-driving car must cope with weather, road layouts and unexpected obstacles; a diagnostic model must stay accurate across diverse patients.
Reliability	**Consistent delivery of accurate outcomes over time**, including under continuous operation. A foundation of trust: unreliable finance models cause large losses; unreliable diagnostic AI endangers patients.

Challenges & mitigation strategies

- **Data quality & diversity** — representative, unbiased sampling and continuous data monitoring.
- **Adversarial attacks** — robust/**adversarial training** and constant evaluation against subtly altered inputs.
- **Transfer learning & generalisation** — adapt to new contexts without extensive retraining.
- **Model monitoring & feedback loops** — observe real-world performance and improve over time.
- **Interpretable AI** and **collaborative ecosystems** (researchers, developers, policymakers, domain experts).

12. Quick Revision & Likely Questions

One-liners

- AI = agents that perceive, learn and act towards goals; today's AI is all **weak**. Strong AI (Searle) = a machine with a mind; not achieved.
- Ethics: **descriptive** (is) • **normative** (ought) • **meta** (nature) • **applied** (fields).
- Normative: **deontology** (duty, Kant) • **consequentialism** (outcomes, utility) • **virtue** (character, Aristotle).
- Ethics ≠ law but overlap; ethics acts as **soft law**.
- Machine ethics hurdles: no feelings, **no agreed theory**, naive utilities misfire → use **test cases** and bias-testing tools.
- Four core principles: **Bias & Fairness • Transparency & Explainability • Privacy & Security • Robustness & Reliability**.

Likely questions

1. Differentiate strong and weak AI with examples. (2)
2. Explain descriptive, normative and meta-ethics. (6)
3. Compare deontological, consequentialist and virtue ethics with one example. (8)
4. Discuss the relationship between ethics and law. (5)
5. Using the toddler–letter example, explain difficulties in machine ethics. (6)
6. How does moral diversity affect testing of ethical AI? Name bias-testing tools. (5)
7. Explain the four core principles of responsible AI. (10)
8. Distinguish transparency from explainability; list challenges. (6)
9. Explain how robustness and reliability can be achieved in AI systems. (6)